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COURSE MATERIAL

Packet Transfer in Networking

CMSC 131 Lab 3

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This laboratory assignment explores endianness concepts through network packet simulation. You will implement a program that demonstrates proper conversion techniques for cross-architecture network communication.

Learning Objectives

By the end of this laboratory exercise, students should be able to:

- Understand little endian (x86) vs big endian (network) byte ordering
- Implement byte-swapping functions using NASM instructions
- Apply proper endianness handling in network protocols
- Use bit manipulation instructions for data conversion
- Apply shifts and masks in bit-level manipulation

Task

Create a program that simulates network packet transmission between different architectures:

1. Prompt the user for a “network packet” with header fields (source IP, destination IP, port numbers, packet length)
2. Display the packet in x86 little endian format
3. Convert the packet to network big endian format for “transmission”
4. Simulate receiving the packet and converting back to little endian

Packet Structure

For example, a network packet should contain these fields:

- **Source IP:** 192.168.1.100 (32-bit in fixed-point representation)
- **Destination IP:** 10.0.0.1 (32-bit in fixed-point representation)
- **Source Port:** 8080 (16-bit in fixed-point representation)
- **Destination Port:** 80 (16-bit in fixed-point representation)
- **Packet Length:** 1024 (32-bit)

Implementation Details

- Display **hex values** to clearly show byte ordering
- For optimal speed, the network should **only store and persist values through registers — this means no reserved memory or storage variables are allowed!**
- For educational purposes, the network should **implement endian conversion through an algorithm of choice, instead of using the `xchg`, `bswap`, or any other macro-like instructions!**
- You may use the `dump_regs` macro to show the values in registers as hex

- Use provided I/O functions for display

Expected Output

```

1  Input the source IP: 1921681000
2  Input the destination IP: 10001
3  Input the source port: 8080
4  Input the destination port: 80
5  Input the packet length in bits: 1024
6  Received Little Endian Packet!
7  Register Dump # 1
8  EAX = 00000400 EBX = 00501F90 ECX = 00000400 EDX = 00001F90
9  ESI = 728A8668 EDI = 00002711 EBP = 007FFEC8 ESP = 007FFEA8
10 EIP = 0031157B FLAGS = 0206                PF
11 Converting to Big Endian...
12 Register Dump # 2
13 EAX = 00040000 EBX = 901F5000 ECX = 00040000 EDX = 0000728A
14 ESI = 68868A72 EDI = 11270000 EBP = 007FFEC8 ESP = 007FFEA8
15 EIP = 0031159A FLAGS = 0216                AF PF
16 Sent Big Endian Packet!
17 .....
18 .....
19 .....
20 Received Big Endian Packet!
21 Converting to Little Endian...
22 Register Dump # 3
23 EAX = 00000400 EBX = 00501F90 ECX = 00000400 EDX = 00006886
24 ESI = 728A8668 EDI = 00002711 EBP = 007FFEC8 ESP = 007FFEA8
25 EIP = 00311618 FLAGS = 0216                AF PF

```

EDX and ECX are unused in the implementation (by preference). However, note the EBP, ESP, EIP, and FLAGS registers are used internally by the network and should not handle packets!

Starter Code

No starter code template is provided for this exercise. *You can do it!*

Testing

Verify your implementation handles:

- Correct round-trip conversion (original = final)
- Proper byte ordering in network format
- Various IP addresses and port combinations

Laboratory Defense

The instructor will be available during scheduled laboratory hours in the assigned room for defense or consultation. You will be catered to at a first-come first-served basis. It is mandatory to present your code, answer inquiries, and perform live programming if the instructor deems it necessary to further check your understanding.

If a laboratory defense for an activity fails to be conducted within **one month** of its assignment, the instructor will be forced to give a failing grade. Scoring will be done during the laboratory defense (no defense, no grade policy). Extensions may be granted due to extraordinary circumstances, but ultimately remains up to the judgment of the instructor.

Academic Honesty

The usage of Large Language Models (e.g. ChatGPT, Claude, Deepseek, etc.) to generate code is considered cheating. As cheating is against the university's code of ethics, it is subject to failure in the course and harsh disciplinary action.

Rubric for Programming Exercises (50 pts)

Criteria (10 Points Each)	Excellent (9-10)	Good (6-8)	Fair (3-5)	Poor (0-2)
Program Correctness	Program executes correctly with no syntax or runtime errors, meets/exceeds specifications, and displays correct output	Program executes and outputs with minor errors, yet meets specifications	Program executes and outputs with major errors, yet somehow meets specifications	Program does not execute or does not meet specs
Logical Design	Program is logically well-designed with excellent structure and flow	Program has slight logic errors that do not significantly affect the results	Program has significant logic errors affecting functionality	Program logic is fundamentally incorrect
Code Mastery	Programmer demonstrates excellent mastery over the program's code	Programmer demonstrates adequate mastery over the program's code	Programmer demonstrates fair mastery over the program's code	Programmer demonstrates poor mastery over the program's code
Engineering Standards	Program is stylistically well designed from an engineering standpoint	Slight inappropriate design choices (i.e., poor variable names, improper indentation)	Severe inappropriate design choices (i.e., code repetition, redundancy)	Program is poorly written
Documentation*	Program is well-documented: comments exist for clarity, not redundancy	Missing one required comment or some redundant comments	Missing two or more required comments or many redundant comments	Most documentation missing or most documentation is redundant

***Remember:** “Code tells you how, comments tell you why.” — Jeff Atwood, co-founder of Stack Overflow and Discourse

See the [laboratory manual](#) for submission requirements.